

William Farlessyost

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SUMMARY

Mechatronics engineer with a Ph.D. and hands-on research experience in control systems, system identification, dynamic modeling, robotics, autonomous systems, sensor fusion, and machine-learning-assisted engineering analysis. Experienced in moving from physical-system requirements to simulation, control logic, model validation, and prototype implementation across process systems, UAVs, manipulators, and mobile robots.

TECHNICAL SKILLS

Controls & Modeling: Automatic control, system identification, nonlinear dynamical systems, reduced-order modeling, mechanistic modeling, SINDy, fuzzy control methods, simulation, model validation

Robotics & Autonomy: ROS, Gazebo, MoveIt!, SLAM, path planning, sensor fusion, fiducial tracking, manipulator control, mobile robotics

Programming: C/C++, MATLAB, Python, R, Bash/Shell, SQL, Git

Engineering Software: Simulink, ASPEN Plus/Dynamics, SolidWorks, AutoCAD, 3D Slicer, BioCro, SimaPro

Machine Learning: TensorFlow, PyTorch, Scikit-learn, CNN, RNN, LSTM, liquid time-constant neural networks, surrogate modeling

ENGINEERING & RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2020 – May 2025

Purdue University, Sustainable Industrial-Natural Coupled Systems Group

West Lafayette, IN

- Applied nonlinear system-identification methods to dynamic chemical-process and biological-system data to derive compact, interpretable models.
- Integrated machine-learning models with mechanistic simulations to improve predictive accuracy while preserving physically meaningful system structure.
- Developed and evaluated surrogate models for coupled natural-industrial systems under changing operating and climate conditions.
- Analyzed robustness, sensitivity, observability, and reduced-sensor configurations to support monitoring and control-oriented model design.
- Implemented computational workflows in Python and MATLAB and communicated engineering tradeoffs through publications, conference presentations, and technical figures.

Undergraduate Research Assistant, Controls

Aug. 2019 – Jan. 2020

University of North Carolina Asheville

Asheville, NC

- Validated and implemented a quadrotor UAV control system for tracking and softly landing on a moving platform.
- Integrated fiducial-tag recognition, sensor fusion, ROS, and Simulink components and tested system behavior across simulated and physical operating conditions.

Mechatronics Senior Design Project

Aug. 2019 – May 2020

NASA Lunabotics Challenge

Asheville, NC

- Designed and built two autonomous double-Ackermann-steered mining robots for excavation and material transport.
- Contributed to mechanical design, electrical integration, SLAM, path planning, autonomous navigation, and closed-loop control.

Undergraduate Research Assistant, Robotics

Aug. 2019 – May 2020

University of North Carolina Asheville, NSF-Funded Research

Asheville, NC

- Developed an apprenticeship-learning system for a UR3e collaborative robot with an RG2 gripper to sort previously unseen objects.
- Implemented motion planning and simulation using ROS, MoveIt!, and Gazebo.

Undergraduate Research Assistant, Robotics

Sept. 2017 – May 2019

University of North Carolina Asheville

Asheville, NC

- Developed a Jenga-playing robot for reinforcement-learning research, including mechanical and electrical design, ROS integration, motion planning, and Gazebo simulation.

ADDITIONAL PROFESSIONAL EXPERIENCE

Founder & AI Product Developer May 2025 – Present
Fee Faux Fum Remote

- Designed and implemented an end-to-end technical product using Python, FastAPI, Supabase, document parsing, structured extraction, rule-based analysis, and automated report generation.
- Managed system architecture, debugging, deployment workflows, user-interface requirements, and iterative product validation for a live beta platform.

AI Trainer, Multimodal AI Evaluation Dec. 2025 – Present
Handshake AI Fellowship Remote

- Evaluated complex technical outputs for correctness, reasoning quality, consistency, and instruction following using structured assessment criteria.

EDUCATION

Purdue University 2025
Ph.D., Agricultural and Biological Engineering
Majors: Computational Science & Engineering; Ecological Sciences & Engineering

University of North Carolina Asheville 2020
B.S.E., Mechatronics Engineering (joint NCSU–UNCA program) *Summa cum laude*
B.A., Applied Mathematics *Summa cum laude*

SELECTED PUBLICATIONS & AWARDS

Farlessyost W, Singh S. “Reduced order dynamical models for complex dynamics in manufacturing and natural systems using machine learning.” *Nonlinear Dynamics*. 2022; 110(2):1613–1631.

Farlessyost W, Singh S. “Improving Mechanistic Model Accuracy with Machine Learning Informed Physics.” *Foundations of Computer Aided Process Design*, 2024.

NSF Graduate Research Fellowship Program 2022 – 2025

Departmental Professionalism and Academic Excellence Award 2020

Top Math Student Award, UNCA Department of Mathematics 2020

SIMIODE Differential Equations Modeling Challenge, Meritorious Award 2019

TEACHING

Graduate and Undergraduate Teaching Assistant Jan. 2017 – May 2021
Led laboratories, help sessions, grading, and project support in automatic control, computer systems, thermodynamics, circuits, statics, and statistical modeling.